Chapter 13: Radiate Animals

Introduction

- Phylum **Cnidaria** and Phylum **Ctenophora** are the two phyla discussed in this chapter.
- They are **diploblastic**, meaning they have two embryonic cell layers: **ectoderm** and **endoderm**.
- In adult diploblasts, the **epidermis** is derived from ectoderm, and the **gastrodermis** (gut lining) is derived from endoderm.
- A new stage of development, **gastrulation**, characterizes diploblasts.
- Adult organisms exhibit **radial** or **biradial symmetry** and are not cephalized.
- **Cnidarians** include sea anemones, jellyfishes, and corals.
- **Ctenophores** are known as comb jellies or sea walnuts.

Phylum Cnidaria

- Phylum **Cnidaria** (Gr. *knide*, nettle) includes over 9000 species.
- They possess specialized cells called **cnidocytes**, which contain organelles called **cnidae**.
- The most common type of cnida is the **nematocyst**.
- Cnidarians are an ancient group, mostly marine, abundant in shallow tropical waters.
- Some live symbiotically with other animals (e.g., hermit crabs) or harbor algal cells (**mutuals**) in their tissues.
- **Reef-building corals** are economically important and form essential habitats.
- Four traditional classes: **Hydrozoa**, **Scyphozoa**, **Cubozoa**, and **Anthozoa**. A fifth class, **Staurozoa**, has been proposed.

Figure 1: A, A hermit crab with its cnidarian mutuals. The host snail shell is blanketed with polyps of the hydrozoan **Hydractinia milleri**. The crab gets some protection from predation by the cnidarians, and the cnidarians get a free ride and bits of food from their host's meals. B, Portion of a colony of **Hydractinia**, showing the types of zooids and the stolon (hydrorhiza) from which they grow.

- **Mutualism** is common in cnidarians, such as hydroids living on snail shells inhabited by hermit crabs (Figure 13.1). The cnidarians protect the crab, while the crab provides transport and food scraps.

Figure 2: Cladogram showing hypothetical relationships of cnidarian classes with some shared derived characters (synapomorphies) indicated. Relationships are according to data of Collins et al. (2006).

- The cladogram (Figure 13.2) shows hypothetical relationships among cnidarian classes, highlighting synapomorphies like the **medusa** stage in Medusozoa and the loss of the polyp in some groups.

Form and Function

- Cnidarians exhibit **dimorphism** (two morphological types): **polyp** and **medusa**.
- Both types share a saclike body plan.

Figure 3: Comparison between polyp and medusa types of individuals.

- The **polyp** is adapted to a sedentary life, with a tubular body and a mouth surrounded by tentacles leading to a blind **gastrovascular cavity** (Figure 13.3).
- The **medusa** is adapted for floating or swimming, essentially an unattached polyp with a widened, bell-shaped body (Figure 13.3).
- **Polyps**:
  - Usually attached by a **pedal disc**.
  - Reproduce asexually by **budding**, **fission**, or **pedal laceration**.
  - Budding can form **clones** or **colonies**.
  - Colonies may exhibit **polymorphism**, with specialized polyps for feeding (**hydranths** or **gastrozooids**) and reproduction (**gonangia**).
- **Medusae**:
  - Usually free-swimming with bell-shaped bodies.
  - Often exhibit **tetramerous symmetry**.

- Have sensory structures: **statocysts** (orientation) and **ocelli** (light reception).
- **Hydromedusae** (Hydrozoa) have a **velum** (shelf-like fold) to increase swimming efficiency.
- **Scyphomedusae** (Scyphozoa) lack a velum.

Life Cycles

- Typical life cycle: Zygote → **Planula larva** (motile) → **Polyp** (sessile) → **Medusa** (sexual, free-swimming).
- Variations exist:
  - **Scyphozoa**: Large medusa, small polyp.
  - **Hydrozoa**: Often colonial polyp, pelagic medusa.
  - **Anthozoa**: Only polyp stage (medusa lost).
  - **Hydra**: Only polyp stage (medusa lost).

Body Wall

- Consists of outer **epidermis** (ectoderm), inner **gastrodermis** (endoderm), and **mesoglea** in between.
- **Mesoglea** is gelatinous; thick in medusae (giving the name “jellyfish”) and Anthozoa, thin in Hydrozoa.
- Cnidarians lack true mesodermally derived muscle cells; instead, they have **epitheliomuscular cells**.

Figure 4: At right, structure of a stinging cell. Center, portion of the body wall of a hydra. Cnidocytes, which contain the nematocysts, arise in the epidermis from interstitial cells.

- The body wall of a **Hydra** (Figure 13.4) shows the arrangement of epidermis, mesoglea, and gastrodermis. **Cnidocytes** containing **nematocysts** are abundant in the epidermis.

Figure 5: Epitheliomuscular and nerve cells in hydra.

- **Epitheliomuscular cells** (Figure 13.5) form most of the epidermis and contain contractile myofibrils, serving as a longitudinal muscle layer.

Cnidocytes

- **Cnidocytes** are unique to Cnidaria, producing organelles called **cnidae**.
- **Nematocysts** are a type of cnida used for prey capture and defense.
- Structure: Capsule containing a coiled thread, covered by an **operculum**.
- Discharge mechanism: High **osmotic pressure** (140 atm) causes water to rush in, increasing **hydrostatic pressure** and forcing the thread out with great velocity.
- Some cnidocytes have a trigger-like **cnidocil** (modified cilium).

Figure 6: A, Several types of cnidae shown after discharge. At bottom are two views of a type that does not impale prey; it recoils like a spring, catching any small part of the prey in the path of the recoiling thread. B, Fired and unfired cnidae from **Corynactis californica**.

- Different types of **cnidae** exist (Figure 13.6): some penetrate and inject toxin (**penetrants**), others recoil and entangle prey (**volvents**), or secrete adhesive (**glutinants**).

Feeding and Digestion

- Polyps are carnivorous, catching prey with tentacles.
- Digestion is initially **extracellular** (enzymes discharged into gastrovascular cavity) and completed **intracellularly** (in gastrodermal cells).
- **Medusae** have a mouth at the end of a **manubrium**.
- **Scyphomedusae** have **oral arms** to capture prey.
- **Anthozoans** (e.g., corals) may supplement nutrition with carbon from symbiotic algae (**zooxanthellae**).

Nerve Net

- Diffuse nervous system: **nerve net** at the base of epidermis and gastrodermis.
- **Bidirectional transmission** at synapses (vesicles on both sides).
- No central nervous system, but nerve rings in medusae and marginal sense organs (**rhopalia**) in Scyphozoa and Cubozoa.

Class Hydrozoa

- Mostly marine and colonial.
- Typical life cycle includes both asexual polyp and sexual medusa.

Hydroid Colonies

- Example: **Obelia**.
- Colony structure: **Hydrorhiza** (rootlike stolon), **hydrocauli** (stalks), **coenosarc** (living tissue), **perisarc** (chitinous sheath).
- Zooids: **Hydranths** (feeding) and **gonangia** (reproductive).
- **Thecate** forms (e.g., **Obelia**) have a protective cup (**hydrotheca**) around the polyp; **athecate** forms lack it.

Figure 7: Life cycle of **Obelia**, showing alternation of polyp (asexual) and medusa (sexual) stages. **Obelia** is a thecate hydroid, its polyps as well as its stems being protected by continuations of the nonliving covering.

- The life cycle of **Obelia** (Figure 13.7) illustrates the alternation between the asexual colonial polyp stage and the sexual free-swimming medusa stage.

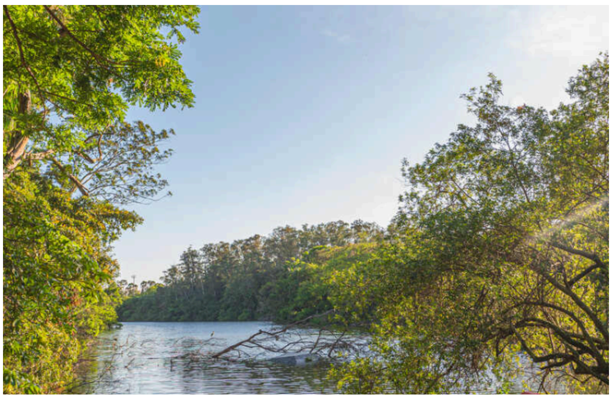

Figure 8: Athecate hydroids. A, **Ectopleura integra**, a solitary polyp with naked hydranths and gonophores. B, **Corymorpha** is a solitary hydroid that produces free-swimming medusae, each with a single trailing tentacle.

- **Athecate hydroids** like **Ectopleura** and **Corymorpha** (Figure 13.8) have naked hydranths without a protective perisarc cup.

Figure 9: In some hydroids, such as this **Tubularia crocea**, medusae are reduced to gonadal tissue and do not detach. These reduced medusae are known as gonophores.

- In some hydroids like **Tubularia** (Figure 13.9), medusae are reduced to **gonophores** that remain attached to the colony.
- **Hydromedusae**:
  - Small, with a **velum**.
  - Mouth at end of **manubrium**.
  - Four **radial canals** and a **ring canal**.
  - Nerve rings, **statocysts**, and **ocelli**.

Figure 10: Bell medusa, **Polyorchis penicillatus**, medusa stage of an unknown attached polyp.

- **Polyorchis penicillatus** (Figure 13.10) is an example of a bell medusa.

Figure 11: Structure of **Gonionemus**. A, Medusa with typical tetramerous arrangement. B, Cutaway view showing morphology. C, Portion of a tentacle with its adhesive pad and ridges of nematocysts. D, Tiny polyp, or hydroid stage, that develops from the planula larva. It can produce more polyps by budding (frustules) or produce medusa buds.

- **Gonionemus** (Figure 13.11) shows the typical structure of a hydromedusa, including the velum, manubrium, radial canals, and nerve rings.

Freshwater Medusae

- **Craspedacusta sowberii**.
- Medusa is dominant; polyp is tiny (**Microhydra**).

Figure 12: Life cycle of **Craspedacusta**, a freshwater hydrozoan. The polyp has three methods of asexual reproduction: by budding off new individuals, which may remain attached to the parent (colony formation); by constricting off nonciliated planula-like larvae (frustules), which can move around and give rise to new polyps; and by producing medusa buds, which develop into sexual jellyfish.

- The life cycle of **Craspedacusta** (Figure 13.12) involves a tiny polyp that reproduces asexually by budding, frustules, and producing medusa buds.

Hydra: A Solitary Freshwater Hydrozoan

- Solitary polyp, no medusa stage.
- Cylindrical body with **basal disc** for attachment and **hypostome** with mouth and tentacles.
- Feeds on small crustaceans (**Daphnia**); feeding response triggered by **glutathione**.

Figure 13: **Hydra** catches an unwary water flea with the nematocysts of its tentacles. This hydra already contains one water flea eaten previously.

- **Hydra** captures prey using nematocysts on its tentacles (Figure 13.13).
- **Reproduction**:
  - Asexual: **Budding**.
  - Sexual: Temporary gonads (testes/ovaries) appear in autumn.

Figure 14: **Hydra** with developing bud and ovary.

- **Hydra** reproduces asexually by budding and sexually by producing ovaries (Figure 13.14) or testes.

Other Hydrozoans

- Order **Siphonophora**: Polymorphic floating colonies (e.g., **Physalia**, Portuguese man-of-war).
- Contain modified polyps (**gastrozooids**, **dactylozooids**, **gonophores**) and a float (**pneumatophore**).

Figure 15: A Portuguese man-of-war colony, **Physalia physalis** (order Siphonophora, class Hydrozoa). Colonies often drift onto southern ocean beaches, where a hazard to bathers. Each colony of medusa and polyp types is integrated to act as one individual. As many as 1000 zooids may be found in one colony. The nematocysts secrete a powerful neurotoxin.

- **Physalia physalis** (Figure 13.15) is a colonial siphonophore with a gas-filled float and specialized zooids for feeding, reproduction, and defense.
- **Hydrocorals**: Secrete calcareous skeletons (e.g., **Millepora**, fire coral).

Figure 16: These hydrozoans form calcareous skeletons that resemble true coral. A, **Stylaster roseus** (order Stylasterina) occurs commonly in caves and crevices in coral reefs. These fragile colonies branch in only a single plane and may be white, pink, purple, red, or red with white tips. B, Species of **Millepora** (order Milleporina) form branching or platelike colonies and often grow over the horny skeleton of gorgonians, as is shown here. They have a generous supply of powerful nematocysts that produce a burning sensation on human skin, justly earning the common name fire coral. The inset photo shows the extended tentacles.

- Hydrocorals like **Stylaster** and **Millepora** (Figure 13.16) resemble true corals but are hydrozoans. **Millepora** is known as fire coral due to its powerful stings.

Class Scyphozoa

- “True” jellyfishes; medusa is the dominant stage.
- Lack a velum.
- Margin of umbrella has **lappets** and **rhopalia** (sense organs).
- **Rhopalia** contain **statocysts** and sometimes **ocelli**.

Figure 17: Giant jellyfish, **Cyanea capillata** (order Semaeostomeae, class Scyphozoa). A North Atlantic species of **Cyanea** reaches a bell diameter exceeding 2 m. It is known as the "sea blubber" by fishermen.

- **Cyanea capillata** (Figure 13.17) is a giant jellyfish that can reach over 2 meters in diameter.
- **Aurelia** (Moon Jelly):
  - Short tentacles, feeds on plankton caught in mucus.
  - Four **gastric pouches** with **gastric filaments** (nematocyst-bearing).
  - Complex system of **radial canals**.

Figure 18: Moon jellyfish **Aurelia aurita** (class Scyphozoa) is cosmopolitan in distribution. It feeds on planktonic organisms caught in mucus on its umbrella.

- **Aurelia aurita** (Figure 13.18) is a common scyphozoan with a characteristic four-part symmetry visible in its gastric pouches.
- **Life Cycle**:
  - Zygote → **Planula** → **Scyphistoma** (hydralike polyp).
  - Scyphistoma undergoes **strobilation** to form a **strobila**.
  - Strobila releases saucer-like **ephyrae**, which grow into medusae.

Figure 19: Life cycle of **Aurelia**, a marine scyphozoan medusa.

- The life cycle of **Aurelia** (Figure 13.19) features the unique process of **strobilation**, where the polyp (**scyphistoma**) produces a stack of young medusae (**ephyrae**).
- **Cassiopeia** (Upside-down jellyfish) and **Rhizostoma**:
  - Lack tentacles on margin.
  - Oral arms fused to form branched canals with many "mouths".
  - **Cassiopeia** harbors **zooxanthellae**.

Class Staurozoa

- **Stauromedusae**.
- No medusa stage; solitary stalked polyp.
- Top of polyp resembles a medusa with eight arms ending in tentacle clusters.

Figure 20: **Thaumatoscyphus hexaradiatus** are an example of class Staurozoa.

- **Thaumatoscyphus hexaradiatus** (Figure 13.20) represents the class Staurozoa, which are stalked polyps resembling attached medusae.

Class Cubozoa

- **Box jellyfishes**.
- Medusa dominant; bell is square in cross-section.
- Tentacles hang from a blade-like **pedalium** at each corner.
- **Rhopalia** with complex eyes (cornea, lens).
- Have a **velarium** (analogous to velum).
- Strong swimmers, voracious predators.
- **Chironex fleckeri** (Sea wasp): Potentially fatal stings.

Figure 21: **Carybdea**, a cubozoan medusa.

- **Carybdea** (Figure 13.21) is a cubozoan medusa with a box-like shape and pedalia at the base of its tentacles.

Class Anthozoa

- “Flower animals”; polyps only, no medusa stage.
- All marine.
- Gastrovascular cavity divided by **septa** (**mesenteries**).
- Three subclasses:
  - **Hexacorallia** (Zoantharia): Hexamerous symmetry (sea anemones, stony corals).
  - **Ceriantipatharia**: Tube anemones, thorny corals.
  - **Octocorallia** (Alcyonaria): Octomerous symmetry (soft and horny corals).

Figure 22: Sea anemones are the familiar and colorful “flower animals” of tide pools, rocks, and pilings of the intertidal zone. Most, however, are subtidal, their beauty seldom revealed to human eyes. These are rose anemones, **Tealia piscivora** (subclass Hexacorallia, class Anthozoa).

- Sea anemones like **Tealia piscivora** (Figure 13.22) are typical anthozoans with a flower-like appearance.

Figure 23: A, Orange sea pen **Ptilosarcus gurneyi** (order Pennatulacea, class Anthozoa). Sea pens are colonial forms that inhabit soft bottoms. The base of the fleshy body of the primary polyp is buried in the bottom. It gives rise to numerous secondary, branching polyps. B, Close-up of a gorgonian. The pinnate tentacles characteristic of subclass Octocorallia are apparent.

- Anthozoans also include colonial forms like the sea pen **Ptilosarcus** and gorgonians (Figure 13.23), which belong to the subclass Octocorallia.

Sea Anemones (Order Actiniaria)

- Cylindrical body with **oral disc** and tentacles.
- **Siphonoglyph**: Ciliated groove in pharynx to create water currents.
- Gastrovascular cavity divided by paired **primary septa**.
- **Septal filaments** with nematocysts; **acontia threads** can be protruded for defense.

Figure 24: Structure of a sea anemone. Free edges of septa and acontia threads are equipped with nematocysts to complete paralyzation of prey begun by the tentacles.

- The internal structure of a sea anemone (Figure 13.24) reveals the pharynx, siphonoglyph, and septa that divide the gastrovascular cavity.

- Feeding behavior is chemically controlled: **Asparagine** activates tentacle bending, and **glutathione** induces swallowing.

- Locomotion: Glide on pedal disc; some can swim (**Stomphia**).

Figure 25: A. A sea anemone that swims. B. When attacked by a predatory sea star **Dermasterias**, the anemone **Stomphia didemon** (subclass Hexacorallia, class Anthozoa) detaches from the bottom and rolls or swims spasmodically to a safer location.

- Some anemones, like **Stomphia** (Figure 13.25), can detach and swim to escape predators like sea stars.
- Mutualism: With hermit crabs, **zooxanthellae**, and anemone fishes (**Amphiprion**).

Figure 26: Orangefin anemone fish (**Amphiprion chrysopterus**) nestles in the tentacles of its sea anemone host. Anemone fishes do not elicit stings from their hosts but may lure unsuspecting other fish to become meals for the anemone.

- Anemone fish (Figure 13.26) live among the tentacles of sea anemones, protected from the nematocysts by a special mucus.

Hexacorallian Corals (Order Scleractinia)

- **Stony corals** or true corals.
- Miniature anemones in calcareous cups (**exoskeleton**).
- No siphonoglyph.
- **Hermatypic** corals build reefs and contain **zooxanthellae**.

Figure 27: A, Cup coral **Tubastrea** sp. Its polyps form clumps resembling groups of sea anemones. Although often found on coral reefs, **Tubastrea** is not a reef-building coral (ahermatypic) and has no symbiotic zooxanthellae in its tissues. B, Polyps of **Montastrea cavernosa** are tightly withdrawn during daytime but open to feed at night, as in C (subclass Hexacorallia).

- Cup corals like **Tubastrea** (Figure 13.27) are solitary or form small clumps, while others like **Montastrea** are colonial reef-builders.

Figure 28: Polyp of a hexacorallian coral (order Scleractinia) showing calcareous cup (exoskeleton), gastrovascular cavity, sclerosepta, septa, and septal filaments.

- A coral polyp (Figure 13.28) sits inside a secreted calcareous cup, with sclerosepta projecting into the polyp's base.

Figure 29: Boulder star coral, **Montastrea annularis**, (subclass Hexacorallia, class Anthozoa). Colonies can grow up to 10 feet (3 m) high.

- Massive colonies like the boulder star coral **Montastrea annularis** (Figure 13.29) are major components of coral reefs.

Tube Anemones and Thorny Corals (Subclass Ceriantipatharia)

- **Tube anemones** (Ceriantharia): Solitary, live in tubes.
- **Thorny/Black corals** (Antipatharia): Colonial, horny skeleton with thorns.

Figure 30: A tube anemone (subclass Ceriantipatharia, order Ceriantharia) extends from its tube at night. Its oral disc bears long tentacles around the margin and short tentacles around the mouth.

- Tube anemones (Figure 13.30) live in secreted tubes buried in sediment.

Figure 31: A, Colony of **Antipathes**, a black or thorny coral (order Antipatharia, subclass Ceriantipatharia, class Anthozoa). Most abundant in deep waters in the tropics, black corals secrete a tough, proteinaceous skeleton that can be worked into jewelry. B, The polyps of Antipatharia have six simple, nonretractile tentacles. The spiny processes in the skeleton are the origin of the common name thorny corals.

- Black corals (Figure 13.31) have a hard, proteinaceous skeleton often used in jewelry.

Octocorallian Corals (Subclass Octocorallia)

- Strict **octomerous symmetry** (8 pinnate tentacles, 8 septa).
- Colonial; polyps connected by **solenia** (gastrodermal tubes) in **coenenchyme** (mesoglea).
- **Endoskeleton** of limy spicules or horny protein.
- Examples: Soft corals, Sea fans (Gorgonians), Sea pens (**Renilla**, **Ptilosarcus**).

Figure 32: Polyps of an octocorallian coral. Note the eight pinnate tentacles, coenenchyme, and solenia. They have an endoskeleton of limy spicules often with a horny protein, which may be in the form of an axial rod.

- Octocorallian polyps (Figure 13.32) are characterized by eight pinnate (feather-like) tentacles and are connected by solenia.

Figure 33: A soft coral, **Dendronephthya** sp. (order Alcyonacea, subclass Octocorallia, class Anthozoa), on a Pacific coral reef. The showy hues of this soft coral vary from pink and yellow to bright red and contribute much color to Indo-Pacific reefs.

- Soft corals like **Dendronephthya** (Figure 13.33) add vibrant colors to coral reefs.

Figure 34: Colonial gorgonian, or horny, corals (order Gorgonacea, subclass Octocorallia, class Anthozoa) are conspicuous components of reef faunas. These examples are from the western Pacific. A, Red gorgonian **Melithaea** sp. B, A sea fan, **Subergorgia mollis**. C, Red whip coral, **Ellisella** sp.

- Gorgonians (Figure 13.34), including sea fans and whip corals, have a tough horny skeleton.

Coral Reefs

- Large formations of calcium carbonate in shallow tropical seas.
- Built primarily by **hermatypic corals** and **coralline algae**.
- Require warmth, light (for **zooxanthellae**), and undiluted seawater.
- Types: **Fringing reefs**, **Barrier reefs**, **Atolls**, **Patch reefs**.
- Zones: Reef front, Reef crest, Reef flat.
- Threats: Nutrient enrichment, overfishing, pesticides, global warming (**coral bleaching**), ocean acidification (makes precipitation of $CaCO_3$ more difficult).

Figure 35: A, Profile of a barrier reef. B, Portion of an atoll from the air. Reef slope plunges into deep water at left (dark blue), lagoon at right.

- Coral reefs form different structures, such as barrier reefs and atolls (Figure 13.35), with distinct zones like the reef front and lagoon.

Phylum Ctenophora

- **Comb jellies**; about 150 species, all marine.
- **Biradial symmetry**.
- Eight rows of **comb plates** (**ctenes**) for locomotion.
- Lack nematocysts (usually); possess **colloblasts** (glue cells) for prey capture.
- **Bioluminescence** is common.

Figure 36: A, Comb jelly **Pleurobrachia** sp. (order Cydippida). Its fragile beauty is especially evident at night when it luminesces from its comb rows. B, **Mnemiopsis** sp. (order Lobata).

- **Pleurobrachia** and **Mnemiopsis** (Figure 13.36) are common ctenophores. **Mnemiopsis** is known for its impact on fisheries in the Black Sea.

Representative Type: **Pleurobrachia**

- Oral pole (mouth) and aboral pole (**statocyst**).
- **Comb plates**: Fused cilia, beat in unison for swimming.
- **Tentacles**: Two long, retractable tentacles with **colloblasts**.
- **Gastrovascular system**: Mouth, pharynx, stomach, branching canals, anal pores.
- **Nervous system**: Nerve net, aboral statocyst coordinates comb beat.
- **Reproduction**: Monoecious; **cydippid larva**.

Figure 37: Comb jelly **Pleurobrachia**, a ctenophore. A, External view. B, Hemisection. C, Colloblast, an adhesive cell characteristic of ctenophores. D, Portion of comb rows showing comb plates, each composed of transverse rows of long fused cilia.

- The structure of **Pleurobrachia** (Figure 13.37) shows the comb rows, tentacles, and internal gastrovascular canals. Colloblasts on the tentacles capture prey.

Figure 38: A cydippid larva.

- Ctenophores develop via a **cydippid larva** (Figure 13.38), which resembles the adult form.

Other Ctenophores

- **Beroe**: No tentacles, feeds on other ctenophores.
- **Cestum** (Venus's girdle): Ribbon-like.
- **Ctenoplana** and **Coeloplana**: Flattened, creeping forms.

Figure 39: Diversity among phylum Ctenophora. A, **Beroe** sp. (order Beroida). B, **Cestum** sp. (order Cestida). C, **Coeloplana** sp. (order Platyctenea).

- Ctenophore diversity (Figure 13.39) includes the thimble-shaped **Beroe**, the ribbon-like **Cestum**, and the flattened, creeping **Coeloplana**.

Phylogeny and Adaptive Diversification

- **Diploblasts**: Cnidaria and Ctenophora have two cell layers and mesoglea.
- Debate exists over whether they are truly diploblastic or triploblastic (due to muscle cells in mesoglea/entocodon).
- **Cnidarian Phylogeny**: Molecular evidence suggests Anthozoa is basal; medusa evolved later in Medusozoa.
- **Myxozoa**, formerly a separate phylum of fish parasites, are now considered highly derived cnidarians.
- **Adaptive Diversification**: Cnidarians are efficient predators and colonial builders; Ctenophores are specialized swimmers with unique adhesive cells.